

Chapter 6

Trajectory Optimization

6.1 From an Impact Task to a Candidate Motion

A trajectory optimizer answers a conditional question: among the motions allowed by a specified model and its constraints, which performs best under a declared objective? For golf, the result might be a candidate downswing that delivers a desired contact state with limited actuator effort. It is neither a measured human strategy nor a guarantee that a player can execute the motion. Those conclusions need additional evidence.

The previous chapters establish the pieces of this question. Configuration geometry determines how motion and force are represented. Underactuated dynamics restrict which timed paths are feasible. Transverse and event analysis explain how perturbations affect a nominal motion and its impact outcome. Optimization joins these pieces by choosing a feasible reference, and possibly a feedback policy, under explicit performance and uncertainty criteria. A reference need not be optimal before it is useful for tracking or identification.

Let x contain the mechanical state and any required actuator or environmental states, and let u be the command. For a smooth phase, a deterministic optimal-control problem can be written

$$\begin{aligned} \min_{x(\cdot), u(\cdot), T} \quad & \phi(x(T), T) + \int_0^T L(x(t), u(t), t) dt, \\ & \dot{x} = f(x, u, t), \quad x(0) = x_0, \\ & c(x, u, t) \leq 0, \quad \psi(x(T), T) = 0, \\ & 0 < T_{\min} \leq T \leq T_{\max}. \end{aligned} \tag{6.1}$$

Bounds on duration are modeling choices, not mandatory features of every OCP; they are useful when unlimited waiting or pumping would undermine the intended task. The initial state, admissible controls, regularity, and event sequence must be specified. Existence of a minimizer also requires suitable compactness/coercivity and closure conditions. A nonempty numerical search space alone does not establish existence or attainability.

A stationary ball can make absolute start time irrelevant in an autonomous model. Duration still changes acceleration, effort, dissipation, noise accumulation, and actuator feasibility. A moving ball, gust, imposed rhythm, or fatigue state introduces further timing dependence. A pause is feasible only if the full state can be held or evolves admissibly during that pause. Free final time is permission to optimize timing, not permission to ignore it.

The terminal constraint must encode the actual contact geometry. A scalar signed gap $h(x, t) = 0$ can describe an impact guard, while additional target conditions describe strike location, face orientation, and other outputs. Not every equality in ψ is an event guard. For first passage, require no earlier contact and the chosen crossing direction; at a regular event, $h_t + h_x f \neq 0$. A terminal equality alone could select a later crossing or an approach from

the wrong side. Finite face and ball geometry, rather than coincidence of idealized centers, should define a physical collision model.

If post-impact ball speed or dispersion is the objective, include the impact map and any subsequent flight model needed to compute that outcome. Pre-impact clubhead speed is a proxy that omits strike location, orientation, restitution, friction, and equipment deformation. Acceleration at the endpoint is likewise not automatically a useful impact objective. Select quantities because the model and evidence connect them to the question being investigated.

6.2 Timing, Effort, and Well-Posed Objectives

Consider the nondimensional integrator $\dot{q} = u$, with $q(0) = 0$, $q(T) = D > 0$, and cost $\frac{1}{2} \int_0^T u^2 dt$. Cauchy–Schwarz gives

$$D^2 = \left(\int_0^T u dt \right)^2 \leq T \int_0^T u^2 dt, \quad J_{\min}(T) = \frac{D^2}{2T}, \quad u_* = D/T. \quad (6.2)$$

With unrestricted T , the infimum is zero as $T \rightarrow \infty$ and is not attained at finite duration. Adding ρT , $\rho > 0$, gives $T_* = D/\sqrt{2\rho}$ if no other constraint is active. This simple example shows why a free-time objective needs scrutiny before choosing a solver.

A quadratic torque penalty is a weighted effort surrogate. It is not mechanical work: actuator power is $u^T \dot{q}_a$ for work-conjugate actuator coordinates, and positive/negative work, heat, activation, and fatigue require their own models. Cost weights have units that make the summed terms commensurate, or the variables should be explicitly nondimensionalized. Changing those scales changes the question unless the weights transform too. Low commanded torque does not by itself ensure low muscle force or repeatability.

For a fixed-duration rest-to-rest double integrator, $\dot{q} = v$, $\dot{v} = u$, the minimum of $\frac{1}{2} \int u^2 dt$ with $(q, v)(0) = (0, 0)$ and $(q, v)(T) = (D, 0)$ is

$$\begin{aligned} s &= t/T, & q_* &= D(3s^2 - 2s^3), \\ v_* &= (6D/T)(s - s^2), & u_* &= (6D/T^2)(1 - 2s), & J_* &= 6D^2/T^3. \end{aligned} \quad (6.3)$$

The stationarity condition makes u affine in time; the two integral endpoint conditions determine its coefficients. The strictly convex quadratic cost and linear constraints make this the unique minimizing control up to changes on sets of measure zero. An acceleration bound $|u| \leq a$ is satisfied by this solution only if $6D/T^2 \leq a$.

Changing the same plant's objective changes its control. Minimum-time rest-to-rest travel with $|u| \leq a$ uses $+a$ followed by $-a$, with $T = 2\sqrt{D/a}$. For monotone first passage to $q = D$ from rest with terminal velocity free, maximum arrival speed under the same acceleration bound is $v(T) = \sqrt{2aD}$, achieved by $u = +a$ throughout and $T = \sqrt{2D/a}$. Indeed, $v(T)^2 = 2 \int_0^D u dq \leq 2aD$. The monotonicity and first-passage assumptions matter. These are calculated particle examples, not golf optimizations. They demonstrate why a torque reversal cannot be inferred without specifying the task and constraints.

The left panel uses the integrator with $D = 1$ and $\rho = 0.5$. The right uses the double integrator with $D = 1$ and $|u| \leq 1$: the effort-minimizing rest-to-rest example has $T = \sqrt{6}$,

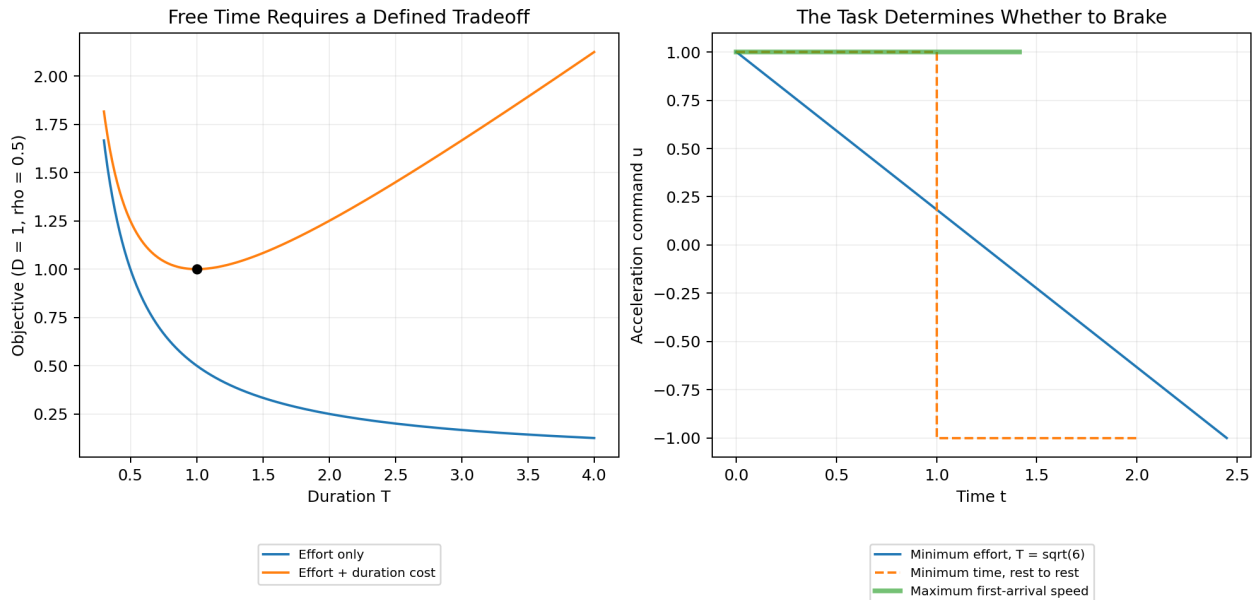


Figure 6.1: Different Objectives Select Different Controls

the minimum-time rest-to-rest example has $T = 2$, and maximum first-arrival speed uses $T = \sqrt{2}$. Their different endpoint and duration requirements explain the different controls.

6.3 Necessary Conditions and Hamiltonian Conventions

On a smooth unconstrained arc of a normal extremal, use the minimization convention

$$\mathcal{H} = L + p^T f, \quad \dot{p} = -\mathcal{H}_x, \quad u_* \in \arg \min_{u \in \mathcal{U}} \mathcal{H}(x_*, u, p, t). \quad (6.4)$$

Here gradients in the costate equation are columns. For regular terminal equalities, there is a multiplier ν such that

$$p(T) = \phi_x + \psi_x^T \nu, \quad \mathcal{H}(T) + \phi_T + \nu^T \psi_T = 0. \quad (6.5)$$

The second condition applies to an interior free final time; it is absent for fixed time and is modified at an active time bound. Endpoint derivatives hold x fixed. State/path constraints require their own multipliers and possibly junction conditions; the displayed unconstrained adjoint is not a complete formula for contact or state-constrained arcs. Abnormal extremals cannot normalize the running-cost multiplier to one and need the general maximum/minimum principle.

The alternative maximum convention uses a consistently transformed costate and Hamiltonian, commonly $\tilde{p}^T f - L$. Maximizing $L + p^T f$ with the minimization costate equations reverses the control selection. For $L = u^2/2$ and $f = u$, minimization gives $u = -p$; maximization over unbounded u has no finite solution. In an autonomous free-time problem without explicit endpoint-time dependence, the normal Hamiltonian is zero along the optimum on the applicable smooth arcs. A fixed-time problem does not inherit that zero condition.

These are necessary conditions under the relevant assumptions, not a general certificate of global optimality. If the Hamiltonian is affine in a bounded scalar input, a nonzero switching coefficient selects a bound. If that coefficient vanishes over an interval, higher-order conditions may determine a singular arc. Quadratic effort, actuator dynamics, rate constraints, or active state constraints change this structure. A smooth wrist release does not identify a singular optimal-control arc or a pattern of muscle activation.

6.4 Transcription and Numerical Verification

Use normalized time $s = t/T \in [0, 1]$. The chain rule and change of variables give

$$\frac{dx}{ds} = T f(x, u, Ts), \quad J = \phi(x(1), T) + T \int_0^1 L(x(s), u(s), Ts) ds. \quad (6.6)$$

Thus a constant physical speed $f = 2$ over $T = 3$ gives $dx/ds = 6$. Multiplying dx/ds by T instead would give the wrong physical duration. Both dynamics and objective must carry the same consistent scaling.

The numerical decision vector contains state samples, control parameters, possibly duration and policy parameters, and any algebraic variables. Write equalities and inequalities separately. Scale residuals by meaningful state, force, and time scales; a solver tolerance is otherwise difficult to interpret. Derivatives must include the dependence on T and on event or reset variables.

6.4.1 Shooting and Collocation

Multiple shooting introduces interval initial states and enforces $x_{k+1} - \widehat{\Phi}_k(x_k, u_k, T) = 0$, where $\widehat{\Phi}_k$ is a numerical flow map for the declared control parameterization. Shorter integrations and sparse continuity constraints often improve conditioning relative to single shooting. They do not remove physical instability, guarantee convergence, or make integration error vanish. Matching the numerical map to solver tolerance is different from matching the exact differential equation.

Direct collocation represents states and controls by chosen polynomials and imposes dynamics at selected points. One concrete Hermite–Simpson interval, of physical duration $h_k = T(s_{k+1} - s_k)$, uses

$$\begin{aligned} x_c &= \frac{1}{2}(x_k + x_{k+1}) + \frac{h_k}{8}(f_k - f_{k+1}), \\ 0 &= x_{k+1} - x_k - \frac{h_k}{6}(f_k + 4f_c + f_{k+1}). \end{aligned} \quad (6.7)$$

Here $f_c = f(x_c, u_c, t_c)$ and, for a linearly interpolated control, $u_c = (u_k + u_{k+1})/2$. A cubic Hermite state interpolant has the endpoint derivatives f_k, f_{k+1} ; its midpoint derivative condition yields the defect above. Use a matching quadrature rule for the cost. Smoothness and interpolation choices determine accuracy; not every collocation method uses these polynomials.

For the rest-to-rest cubic example with $D = 1, T = 2$, the endpoint states are $(0, 0)$ and $(1, 0)$, and accelerations are 1.5 and -1.5 . These formulas give $x_c = (0.5, 0.75)$ and

zero Simpson defect. That exact polynomial check is a useful unit test before applying transcription to multibody dynamics.

6.4.2 What Solver Success Does Not Establish

Dynamics enforced at finitely many points need not hold between them. For example, a residual proportional to $s(s - 1/2)(s - 1)$ vanishes at three sample points and is nonzero elsewhere. A path $q(s) = 4s(1 - s)$ satisfies $q \leq 1/2$ at its endpoints and violates it at the midpoint. Checking a few additional points reduces the risk of missing such behavior but is not a continuous-interval proof.

Reintegrate the optimized control with an independent, tighter error-controlled solver, preserving its actual interpolation and the same initial state. Compare the resulting path, first event, terminal outputs, and constraints with the transcription. Inspect differential residuals between nodes and repeat on refined meshes until the quantities of interest stabilize to stated tolerances. Integration agreement supports numerical fidelity to the model; it does not validate the model itself. A rigorous continuous-time certificate requires additional bounds on residuals, interpolation, and constraint margins.

Nonconvex solvers may find different feasible local solutions from different initial guesses, or return an infeasible stationary point. Report termination status, primal residuals, stationarity measures, active bounds, discretization, and results across starts. Use a finite stopping budget and return failure when acceptance criteria are unmet. Relaxing a constraint for continuation is useful only if the original constraint is restored and checked before accepting the result. No algorithmic workflow should promise a feasible trajectory for an infeasible problem.

6.5 Mechanics and Clubhead Speed

For minimal mechanical coordinates q , write

$$\dot{q} = v, \quad M(q)\dot{v} + b(q, v) = B(q)u + F_{\text{ext}}(q, v, t). \quad (6.8)$$

The bias b includes the chosen velocity, potential, and passive terms once. The state is $x = (q, v)$, possibly augmented; M acts on configuration accelerations, not on a second derivative of the entire state. On a regular patch, a left annihilator $N_B B = 0$ gives $N_B(M\dot{v} + b - F_{\text{ext}}) = 0$. This tests membership in the unrestricted actuation image. Input magnitude/rate limits and actuator states remain separate constraints. A correct first-order transcription already enforces these passive equations, so adding a redundant annihilator constraint is not generally necessary.

For closed chains or maintained contacts, include the independent closure constraints, compatible velocities, and constraint reactions or use a valid minimal-coordinate reduction. Ideal reactions can exert force without doing work along an admissible velocity. Unilateral contact also requires the appropriate nonpenetration, reaction-sign, friction, and mode conditions. An optimizer that ignores these conditions can obtain an attractive objective from a mechanically impossible motion. Contact switches need separate phases and reset/guard

constraints, or a contact formulation whose approximation and convergence are explicitly assessed.

If the clubhead point is $r(q)$, its velocity is $J_r(q)v$ and its speed is $\|J_r v\|$. With the absolute planar angles of Chapter 5,

$$\|\dot{r}\|^2 = l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2l_1 l_2 \cos(\theta_1 - \theta_2) \dot{\theta}_1 \dot{\theta}_2. \quad (6.9)$$

For aligned unit links and $\dot{\theta}_2 = 1$, changing $\dot{\theta}_1$ from -1 to 1 changes tip speed from zero to two. Distal angular velocity alone therefore does not determine linear clubhead speed. A moving shoulder, distributed shaft deformation, and a different angle convention require the corresponding kinematics rather than this particular two-link formula.

The passive club equation relates arm acceleration, coupling, gravity, and velocity terms. It does not specify the sign of shoulder torque or an optimal switching schedule. A credible golf optimization must publish its parameters, initial state, objective, input bounds, contact conditions, and converged solution before describing such a schedule as a result. Compare alternative objectives and model assumptions, and inspect whole-system and segment energy/power balances to explain the mechanism. A nominal simulation is not evidence that a player should actively brake a segment.

6.6 Repeatability Requires an Uncertainty Model

Separate initial-state variation, uncertain parameters, disturbances, motor command variability, observation noise, and feedback/estimation errors. They enter different equations and can have different time correlations. A single Gaussian random vector can model a random initial condition or a finite set of uncertain parameters. It is not automatically a continuous noise process, nor signal-dependent noise merely because its covariance is named in the objective.

Harris and Wolpert's 1998 theory investigated eye and goal-directed arm movements using signal-dependent neural-command noise. It motivates a class of hypotheses; it does not supply a universal golf joint-torque noise law. Torque is already the result of activation and muscle mechanics. A chosen model such as conditional command covariance proportional to u_j^2 needs calibration, units, sampling interval, correlation structure, and validation for the intended movement.

For a local discrete linear model with feedback $u = u_* - K_k \delta x$, independent zero-mean noise w_k of identity covariance, and $\delta x_{k+1} = A_{cl,k} \delta x_k + G_k w_k$, covariance satisfies

$$A_{cl,k} = A_k - B_k K_k, \quad \Sigma_{k+1} = A_{cl,k} \Sigma_k A_{cl,k}^T + G_k G_k^T. \quad (6.10)$$

This is exact for the stated linear model. Substituting Jacobians and nominal diffusion amplitudes from nonlinear dynamics gives a local approximation. Covariance propagation alone is analysis; covariance steering additionally chooses admissible controls or policies to meet distributional objectives or constraints. Neither phrase identifies a biological controller.

Freezing signal-dependent diffusion at its nominal value can omit multiplicative terms. For the zero-mean scalar example $\xi_{k+1} = a\xi_k + n\xi_k w_k + gw_k$, with w_k independent of ξ_k ,

$$\Sigma_{k+1} = (a^2 + n^2)\Sigma_k + g^2. \quad (6.11)$$

The $n^2\Sigma_k$ term survives even though the state mean is zero. With $a = 0.5, n = 0.2, g = 0.1, \Sigma_k = 1$, the next variance is 0.30, compared with 0.26 if multiplicative noise is discarded. Correlated or colored noise requires the corresponding cross-covariances or augmented dynamics. State and diffusion linearizations must therefore be described, not silently replaced with an additive-noise formula.

6.7 Impact Loss, Sensitivity, and Risk

Let y be a physically scaled impact outcome, not just the terminal guard residual. For a random output with mean μ_y and covariance Σ_y , a quadratic loss satisfies

$$\mathbb{E}[(y - y_d)^T W (y - y_d)] = (\mu_y - y_d)^T W (\mu_y - y_d) + \text{tr}(W \Sigma_y), \quad W \succeq 0. \quad (6.12)$$

This identity only needs finite second moments. Penalizing covariance alone can favor a repeatable miss. Combining mean outcome, dispersion, effort, and constraint risk with declared weights makes the tradeoffs explicit. For a nonquadratic expected cost, the mean trajectory alone generally does not determine the expectation.

At an event, Chapter 4 gives the output sensitivity to a fixed-time state perturbation just before the nominal event:

$$C_e = H_x - \frac{(H_x f + H_t)h_x}{h_t + h_x f}, \quad \Sigma_y \approx C_e \Sigma^- C_e^T. \quad (6.13)$$

Here $y = H(x, t)$ is a smooth pre-event output and $h = 0$ is the regular guard. The approximation assumes small perturbations and the same event sequence; parameter/measurement uncertainty and cross-covariances add further terms. If post-impact states are compared at a common time, use the reset and saltation map from Chapter 4. Near grazing or changing contact order, these local derivatives can fail.

For an actual hitting event, $h(x(t_e), t_e) = 0$ by definition for trajectories that hit it. Its variance does not measure all strike errors, and trajectories that never hit require a separate failure outcome. Propagate the relevant face, location, velocity, and ball-flight outputs instead. The condition number of ψ_x is also not an absolute error-amplification measure: I and $1000I$ both have condition number one but their gains differ by a factor of one thousand. The dynamics-to-event map, physical metrics, input uncertainty, and singular values determine local output sensitivity.

A Gaussian scalar approximation to a one-sided chance constraint $a^T x \leq b$ gives

$$a^T \mu + \Phi_N^{-1}(1 - \epsilon) \sqrt{a^T \Sigma a} \leq b. \quad (6.14)$$

Here Φ_N is the standard normal distribution function. This is a marginal probability statement under the specified distribution, not a guarantee for every state or every time. Ten independent constraints each met with probability 0.95 have joint success probability 0.95^{10} , about 0.599. Allocating failure probabilities whose sum is at most a target ϵ gives a conservative union-bound guarantee for the finite collection, without requiring independence. Continuous-time safety still needs treatment between the checked times. Gaussian models have unbounded tails and cannot provide deterministic hard bounds on every realization.

Local covariance analysis is not a robustness certificate for the nonlinear swing. Validate predicted means, tails, event failures, and constraint violations with independent simulations and held-out measurements, reporting sampling uncertainty. Worst-case bounded-disturbance verification answers a different question and requires the set-invariance conditions of the funnel chapter. A stochastic objective may favor lower effort, a different reference, or stronger active feedback; it need not select passive stability, submaximal speed, or a geometrically wider funnel.

6.8 Search, Comparison, and Scientific Interpretation

Derivative-free population methods can explore policy parameters and provide warm starts when derivatives are unavailable or unreliable. Gradient-based methods can exploit sparse derivatives for local refinement. Neither a population search nor multiple initial guesses guarantees a global solution to a nonconvex swing problem. Contact events do not make derivatives universally unavailable: regular event sensitivities may be computed, while mode changes and grazing require special care. State the actual derivative and contact treatment.

For an equality-constrained task, rejecting every inexact candidate may leave no feasible population. Use a feasibility-preserving parameterization, constraint restoration, or a documented tolerance, and return an explicit failure if no acceptable candidate exists. Penalties and relaxed constraints must not silently redefine the final task. Report computational effort and best feasible objectives, not an unqualified claim of global discovery.

A useful golf comparison holds the task and admissible mechanics fixed while changing one proposed explanation: input distribution, timing, feedback, equipment parameters, or uncertainty model. Check whether the result persists across parameter uncertainty, meshes, initial guesses, and held-out trials. If several objectives reproduce the same nominal motion, observation alone does not identify which objective a human optimizes. Forward interventions and measurable impact predictions can distinguish them.

The output of this chapter is therefore a documented candidate trajectory and, when optimized, a realizable policy with quantified numerical and statistical limits. Its geometry, power flow, transient response, and impact distribution should form one consistent argument. The next chapter asks which neighborhood can be verified for the chosen nonlinear model and controller; that neighborhood cannot be assumed from optimizer success.

6.9 Further Reading

Tedrake’s trajectory-optimization notes compare shooting and transcription. Kelly’s transcription tutorial explains numerical and hybrid formulations. Liberzon’s optimal-control text develops the maximum principle, transversality, existence, and the distinction between necessary and sufficient conditions; translate its Hamiltonian convention consistently. Tedrake’s robust/stochastic notes distinguish probabilistic and bounded-disturbance problems. Harris and Wolpert’s original noise-model paper provides the specific motor-planning hypothesis discussed above, with its original movement scope.

6.10 Exercises

1. Derive both normalized-time equations. Test them on constant speed over a duration different from one. State the units of every cost weight.
2. Verify the free-time integrator infimum, then add duration cost and an input bound. Identify when the bound changes the optimal duration.
3. Derive the rest-to-rest cubic and its effort. Compare its control with the two bounded-acceleration objectives in the text.
4. Derive the normal PMP conditions using the stated minimum convention. Explain what changes at fixed terminal time and active path constraints.
5. Check the Hermite–Simpson cubic example, then exhibit an inter-node constraint violation despite feasible nodes. Propose a verification protocol.
6. Derive clubhead speed in absolute and relative joint angles, including a moving shoulder. Explain why distal angular speed is insufficient.
7. Compute the multiplicative-noise variance example by averaging over independent values $\xi, w \in \{-1, 1\}$. Compare with the additive approximation.
8. Derive the quadratic output-loss identity and event covariance map. Explain why variance of an exactly satisfied hitting guard is inadequate.
9. Compare marginal and joint chance constraints. State which assumptions would allow a statistical claim about continuous-time constraint violations.
10. Specify a reproducible golf optimization and a forward intervention that could falsify its proposed mechanical explanation. Separate numerical convergence, model fidelity, repeatability, and human-policy identification.